• PYTHON •

1. Print

```
print("Hello Bangladesh")
```

```
Hello Bangladesh
```

```
print("I am Kazi Tajkir Hossen")
```

```
I am Kazi Tajkir Hossen
```

```
print("I'm learning Python")
```

```
I'm learning Python
```

```
print("Step Down VC")
```

```
Step Down VC
```

```
print("Semester : 3-1")
print("Section  : C-1")
```

```
Semester : 3-1
Section  : C-1
```

2. Comments

```
# Single Line Comment

print("Google")
```

```
Google
```

```
print(34 + 56)        # Adding Two Numbers
```

```
90
```

```
Z = 45                # Storing value in Z

print(Z)
```

```
45
```

```
"""
This is a Multi-line comment
Written using Tripple quotes
"""
print("Google Colab")
```

```
Google Colab
```

```
"""
Name: Python Practice
Level: Beginner
"""
Y = 50
print(Y)
```

```
50
```

3. Variables & Data Types

```
T = 14
print(T)

K = 21.14
print(K)

Name = 'Tajkir'
print(Name)
```

```
14
21.14
Tajkir
```

```
L = 6            #int
M = 3.14         #float
N = "QWERTY"     #string
O = True         #boolean

print(type(L))
print(type(M))
print(type(N))
print(type(O))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

```
#1. Int to Float

Num = 30
Num_float = float(Num)

print(Num_float)

#2. Float to Int

Num = 45.67
Num_int = int(Num)

print(Num_int)

#3. Int to String

Num = 60
Num_str = str(Num)

print(Num_str)
print(type(Num_str))

#4. String to Int

Num = "60"
Num_int = int(Num)

print(Num_int)
print(type(Num_int))

#5. Float to String

Num = 70.25
Num_str = str(Num)

print(Num_str)
print(type(Num_str))
```

```
#6. String to Float

Num = "70.25"
Num_float = float(Num)

print(Num_float)
print(type(Num_float))
```

```
30.0
45
60
<class 'str'>
60
<class 'int'>
70.25
<class 'str'>
70.25
<class 'float'>
```

4. String

```
B = "BANGLADESH"
print(B)
```

```
BANGLADESH
```

```
print("Kazi " + "Tajkir Hossen")
```

```
Kazi Tajkir Hossen
```

```
W = "World Bank"
print(W[5])
```

```
P = "Python"
print(P[-1])
```

```
n
```

```
G = "Google Map"
print(len(G))
```

```
10
```

5. Operations

```
#Arithmetic

#1
A = 11600560
B = 11600561

print("Addition : ", A + B)

#2
print("Subtraction : ", 2026 - 2003)

#3
print("Multiplication : ", 14 * 21)

#4
M = 55
N = 8

print("Division : ", M / N)
```

```
#5
print("Modulus : ", 55 % 8)
```

```
Addition :  23201121
Subtraction :  23
Multiplication :  294
Division :  6.875
Modulus :  7
```

```
#Comparison

print(14 > 13)         #1
print(21 < 14)         #2
print(55.3 == 55.4)    #3
print(30 != 32)        #4
print(56 >= 65)        #5
```

```
True
False
False
True
False
```

```
#Logoical

T = True
F = False

print(T and F)
print(T or F)
print(not F)
```

```
False
True
True
```

5. User Input

```
Name = input("Enter Name : ")
print(Name)
```

```
Enter Name : Tajkir
Tajkir
```

```
Num = int(input("Enter Number : "))
print(Num)
```

```
Enter Number : 56
56
```

```
Num = float(input("Enter Number : "))
print(Num)
```

```
Enter Number : 3.14159
3.14159
```

```
Hobby = input("Enter Hobby : ")
print("Hobby : ", Hobby)
```

```
Enter Hobby : Cricket
Hobby :  Cricket
```

```
City = input("Enter City Name : ")
print("City : ", City)
```

```
Enter City Name : Chandpur
City :  Chandpur
```

Control Flow

```
Age = 18

if Age >= 18 :
    print("Eligible to VOTE")
```

```
Eligible to VOTE
```

```
Num = int(input("Enter Number : "))

if Num > 0 :
    print("Positive")
else:
    print("Negative")
```

```
Enter Number : 21
Positive
```

```
T = 14

if T % 2 == 0 :
    print("Even")
else :
    print("Odd")
```

```
Even
```

```
Marks = 76

if Marks >= 80 :
    print("Grade : A+")
elif Marks >= 70 :
    print("Grade : A")
elif Marks >= 60 :
    print("Grade : A-")
elif Marks >= 50 :
    print("Grade : B")
else :
    print("Fail")
```

```
Grade : A
```

```
Signal = input("Enter Traffic Signal Color : ")

if Signal == "Red" :
    print("Stop")
elif Signal == "Yellow" :
    print("Get Ready")
elif Signal == "Green" :
    print("Go")
else :
    print("Invalid Signal")
```

```
Enter Traffic Signal Color : Green
Go
```

Loop

```
for i in range(5) :
    print(i)
```

```
0
1
2
3
4
```

```
for i in range(1, 6) :
    print(i)
```

```
1
2
3
4
5
```

```
for i in range(1, 7, 2) :
    print(i)
```

```
1
3
5
```

```
Count = 1

while Count <= 4 :
  print(Count)
  Count += 1
```

```
1
2
3
4
```

```
#1
for i in range(1, 11) :
    if i == 4 :
        continue
    if i == 8 :
        break

    print(i)

print("\n")

#2
for i in range(1, 11) :
    if i == 4 :
        break
    if i == 8 :
        continue

    print(i)
```

```
1
2
3
5
6
7


1
2
3
```

```
for i in range(1, 4) :
    for j in range(1, 4) :
```

```
        print(i, j)
```

```
1 1
1 2
1 3
2 1
2 2
2 3
3 1
3 2
3 3
```

List

```
Num = [1, 2, 3]

print(Num)
```

```
[1, 2, 3]
```

```
Alpha = [1, 3, 5, 7]
print(Alpha[0])

Beta = [2, 4, 6, 8]
print(Beta[-2])
```

```
1
6
```

```
Gamma = [11, 16, 17, 44]
Gamma.append(63)

print(Gamma)
```

```
[11, 16, 17, 44, 63]
```

```
Delta = [23, 20, 11, 21]

print(len(Delta))
```

```
4
```

```
Theta = [12, 34, 56, 78, 90]

print(sum(Theta))
```

```
270
```

Functions

```
def Hello() :
  print("Hello Python")

Hello()
```

```
Hello Python
```

```
def Greet(Name) :
  print("Good Afternoon,", Name)

Greet("Gemini")
```

```
Good Afternoon, Gemini
```

```
def add(A, B):
    return A + B
```

```
print(add(14, 3.14))
```

```
17.14
```

```
def Square(I) :
    return I * I

print(Square(23))
```

```
529
```

```
def Even(N) :

  return N % 6 == 0

print(Even(18))         #1
print(Even(21))         #2
```

```
True
False
```